Fiscal Year:	FY 2019	**Task Last Updated:**	FY 04/14/2020
PI Name:	Uddin, Sardar Ph.D.		
Project Title:	Effects of Simulated Microgravity on Osteoblast-Osteoclast Coupling and Therapeutic Countermeasures		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	11794-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	07/27/2018	**End Date:**	02/10/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/10/2020 due to new PI Dr. David Komatsu taking over the project from Dr. Uddin; grant number remains the same. Information per F. Hernandez/ARC (Ed., 4/14/2020)		
Key Personnel Changes/Previous PI:	NOTE 4/14/2020: End date changed to 2/10/2020 due to new PI Dr. David Komatsu taking over the project from Dr. Uddin. Information per F. Hernandez/ARC.		
COI Name (Institution):	Komatsu, David Ph.D. (State University of New York, Stony Brook)		
Grant/Contract No.:	80NSSC18K1479		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Objective: Astronauts lose 1-1.5% of their bone mass per month during low Earth orbit missions. Spaceflight profiles of 2 years or longer, as projected for Mars missions, will significantly increase bone fragility, potentially compromising the ability of astronauts to perform their mission tasks in transit and on Mars. To facilitate human exploration of Mars, it is therefore essential to understand the underlying causes of microgravity-induced bone loss and develop effective countermeasures to sustain bone structure and mechanical integrity under microgravity conditions. Our long-term goal is to protect the skeletal health of astronauts in space environments so they can perform their mission operations. The objective of the current proposal is to determine the effects of microgravity on osteoblast-osteoclast coupling and communication in order to identity a countermeasure for microgravity-induced decoupling. Our central hypothesis is that simulated microgravity will impair osteoblast-osteoclast coupling and that bisphosphonate (alendronate) and/or anti-RANKL antibody (denosumab) treatment will mitigate microgravity-induced osteoblast-osteoclast decoupling. The rational for this proposal is the fact that microgravity promotes osteoclastic bone resorption and impairs osteoblastic bone formation. It is therefore expected that physiological osteoblast-osteoclast coupling is impaired and that denosumab, a RANKL inhibitor will restore osteoblast-osteoclast coupling and mitigate microgravity-induced bone loss to a greater degree than alendronate, an anti-catabolic agent. Specific Aims and Methods: The central hypothesis will be tested by completing following two specific aims. 1. Determine the effects of simulated microgravity on osteoblast-osteoclast coupling and communication utilizing in vitro co-cultures. Our working hypothesis is that microgravity will alter osteoblast-osteoclast communication processes. Pathway analyses will be conducted by assessing the differential expression of mRNA (e.g., RNAseq, qPCR) and protein (i.e., ELISA, Western blots), as well as histology and histomorphometry. Combined Confocal Raman Scanning Electron Microscopy (CRSEM) will be used to determine functional impairment of bone resorption and formation. 2. Assess the efficacy of denosumab relative to alendronate as a countermeasure to microgravity-induced osteoblast-osteoclast decoupling in vitro. Our working hypothesis is that denosumab will be more efficacious in restoring microgravity-induced impairment of osteoblast-osteoclast coupling as it functions by blocking excess RANKL, while alendronate only affects resorption. Osteoblast-osteoclast communication will be assessed using proteomic and genetic analysis and function restoration will be assessed using histology, histomorphometry, and CRSEM. Significance and Outcomes: Bone remodeling is a complex process that is tightly regulated by osteoblast-osteoclast coupling and communication. Microgravity impairs this process by increasing osteoclast associated resorption and decreasing osteoblastic bone formation. In order to continue to expand our manned spaceflight program, it is essential to determine the effects of microgravity on osteoblast-osteoclast coupling and develop targeted countermeasures to sustain bone health in microgravity. At the completion of the study we expect to characterize the effects of microgravity on osteoblast-osteoclast coupling, identify communication pathways associated with microgravity, as well as establish the efficacy of alendronate and/or denosumab as potential therapeutics to restore osteoblast-osteoclast communication and function. The outcomes of this study will fill NASA's recognized knowledge gap and guiding question in microgravity effects on osteoblast-osteoclast communication and evaluate potential therapeutics to aberrant the microgravity-induced bone loss in astronauts (AH2, 3, CMB1, 2, 6).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research is highly innovative as it for the first time uses creative osteoblast-osteoblast communication on the bone chip to study, bone formation, and resorption. The study will use innovative experimental approaches to study the bone formation and how bone resorption is resulting in the first comprehensive invitro hylomorphic analysis of bone formation and resorption relative to it with changing gravity and Osteoblast-Osteoclast activity. The successful completion of research will allow us to develop new approaches to studies osteoblast-osteoclast communication in different gravitational and pathological conditions. Studying the changes in osteoblast-osteoclast coupling will enable us to identify new therapeutic targets to facilitate treat bone loss in space and earth base conditions.
Task Progress:	Report dated June 2019: We have developed in vitro set up to achieve complete rotation of bone chips in Synthcon culture system. Currently, we are working on growing osteoblast on chips and studying effects of microgravity on bone formation followed by osteoblast-osteoclast co-cultures. NOTE 4/14/2020: End date changed to 2/10/2020 due to new PI Dr. David Komatsu taking over the project from Dr. Uddin. See "Effects of Simulated Microgravity on Osteoblast-Osteoclast Coupling and Therapeutic Countermeasures--PI Komtasu" for subsequent reporting.
Bibliography Type:	Description: (Last Updated:)